
CONTACT INFORMATION	<i>Email:</i> spencerjones@tamu.edu <i>Website:</i> http://cspencerjones.github.io/
EDUCATION	Scripps Institution of Oceanography, University of California, San Diego PhD in Physical Oceanography. Dissertation title: The Global Meridional Overturning Circulation in an idealized two-basin model Advisor: Professor Paola Cessi 2011–2018 Oxford University MPhys in Physics, Second Class Upper Division 2007–2011
POSITIONS	Assistant Professor, Department of Oceanography, Texas A&M University, September 2023–present Visiting Assistant Professor & ACES Fellow, Department of Oceanography, Texas A&M University, August 2021–August 2023 Associate Research Scientist, Lamont-Doherty Earth Observatory, Columbia University, September 2020–July 2021 Postdoctoral Research Fellow, Lamont-Doherty Earth Observatory, Columbia University, June 2018–September 2020
JOURNAL ARTICLES	<i>An asterisk (*) before the author name indicates that Jones was the primary mentor of this author.</i> Wang, Y., Lyu, J., Monkman, T., Jones, S., Pedersen, C., & Balwada, D. (2026). A Multi-scale probabilistic machine learning model for balanced and unbalanced sea surface height decomposition. <i>Journal of Geophysical Research: Machine Learning and Computation</i>, 3(3), e2025JH001186. Roquet, F., Bell, M. J., de Boer, A. M., Ferreira, D., Jones, C. S., LaCasce, J. H., de Lavergne, C., Marshall, D. P. , Munday, D. R., Nycander, J. & Ödalen, M. (2025). Controls of the global overturning circulation of the ocean. <i>npj Climate and Atmospheric Science</i>, 8(1), 304. *Neumann, N. K., & Jones, C. S. (2025). Effects of wind and isopycnal mixing on Southern Ocean surface buoyancy flux and Antarctic Bottom Water formation. <i>Geophysical Research Letters</i>, 52(7), e2024GL112133. Schlichting, D., Hetland, R., & Jones, C. S. (2024). Numerical mixing suppresses submesoscale baroclinic instabilities over sloping bathymetry. <i>Journal of Advances in Modeling Earth Systems</i>, 16(12), e2024MS004321.

Jones, C. S., *Jiang, S., & Abernathey, R. P. (2024). A comparison of diagnostics for AMOC heat transport applied to the CESM large ensemble. *Journal of Advances in Modeling Earth Systems*, 16(8), e2023MS003978.

Xiao, Q., Balwada, D., **Jones, C. S.**, Herrero-González, M., Smith, K. S., & Abernathey, R. (2023). Reconstruction of surface kinematics from sea surface height using neural networks. *Journal of Advances in Modeling Earth Systems*, 15(10), e2023MS003709.

Jones, C. S., Xiao, Q., Abernathey, R. P., & Smith, K. S. (2023). Using Lagrangian filtering to remove waves from the ocean surface velocity field. *Journal of Advances in Modeling Earth Systems*, 15(4), e2022MS003220.

Bailey, S. T., **Jones, C. S.**, Abernathey, R. P., Gordon, A. L., & Yuan, X. (2023). Water mass transformation variability in the Weddell Sea in ocean reanalyses. *Ocean Science*, 19(2), 381-402.

Pavia, F. J., **Jones, C. S.**, & Hines, S. K. (2022). Geometry of the Meridional Overturning Circulation at the Last Glacial Maximum. *Journal of Climate*, 35(17), 5465-5482. (All authors contributed equally to this manuscript)

Jones, C. S., & Abernathey, R. P. (2021). Modeling water-mass distributions in the modern and LGM ocean: Circulation change and isopycnal and diapycnal mixing. *Journal of Physical Oceanography*, 51(5), 1523-1538.

Jones, C. S., & Abernathey, R. P. (2019). Isopycnal mixing controls deep ocean ventilation. *Geophysical Research Letters*, 46(22), 13144-13151.

Jones, C. S., & Cessi, P. (2018). Components of upper-ocean salt transport by the gyres and the Meridional Overturning Circulation. *Journal of Physical Oceanography*, 48(10), 2445-2456.

Jones, C. S., & Cessi, P. (2017). Size matters: Another reason why the Atlantic is saltier than the Pacific. *Journal of Physical Oceanography*, 47(11), 2843-2859.

Cessi, P., & **Jones, C. S.** (2017). Warm-route versus cold-route interbasin exchange in the Meridional Overturning Circulation. *Journal of Physical Oceanography*, 47(8), 1981-1997.

Jones, C. S., & Cessi, P. (2016). Interbasin transport of the Meridional Overturning Circulation. *Journal of Physical Oceanography*, 46(4), 1157-1169.

Jones, C. S., Cenedese, C., Chassignet, E. P., Linden, P. F., & Sutherland, B. R. (2015). Gravity current propagation up a valley. *Journal of Fluid Mechanics*, 762, 417-434.

SELECTED
PREPRINTS

*Bhattacharjee, A., **Jones, S.**, Balwada, D., Elipot, S., & Gutierrez-Villanueva, M. (2026). Estimating the Kinetic Energy Spectrum from the Second-Order Velocity Structure Function using a Regularized Fitting Approach. arXiv preprint arXiv:2604.27200.

FELLOWSHIPS	ACES Fellowship Two-year faculty fellowship, August 2021 - August 2023 Texas A&M University, College Station, TX
	Lamont Postdoctoral Fellowship Two-year postdoctoral fellowship, June 2018 - June 2020 Lamont-Doherty Earth Observatory, Columbia University, Palisades, NY
	Geophysical Fluid Dynamics Fellowship Summer Program, June - August 2013 Woods Hole Oceanographic Institution, Woods Hole, MA
EXTERNAL FUNDING	Collaborative Research: Tracing Carbon and Heat Pathways in the Southern Ocean (PI) Lead PI: Nicole Lovenduski Funding agency: NSF Physical Oceanography Award amount: \$299,284 (Texas A&M portion) Dates: 09/01/2026 - 08/31/2029
	TRIAD: Improving Eddy Parameterization in the MPAS-Ocean Model using Machine Learning (PI) Funding Agency: Department of Energy via TRIAD agreement with Texas A&M University Award amount: \$48,000 Dates: 05/01/2026 - 11/01/2028
	Leveraging Machine Learning, Realistic Simulations, And In-Situ Observations To Infer Submesoscale Transport From SWOT (Co-I) PI: Shafer Smith. Other Co-Is: Dhruv Balwada, Abigail Bodner, Leah Johnson Funding Agency: NASA SWOT Award amount: \$150,000 - \$250,000 (exact amount allocated to Texas A&M subject to change) Dates: 06/04/2024 - 06/04/2028
	Collaborative Research: A global census of submesoscale energetics using in-situ drifter observations and a high resolution ocean model (lead PI) Co-PIs: Dhruv Balwada and Shane Elipot Funding agency: NSF Physical Oceanography Award amount: \$229,234 (Texas A&M portion) Dates: 06/01/2023 - 05/31/2027
	Ventilation and mixing of surface and intermediate waters in the tropical Atlantic: perspectives from Lagrangian particles and tracers. (PI) Funding Agency: NSF Physical Oceanography Award amount: \$256,224 Dates: 01/01/2023 - 12/31/2026

TEACHING AT
TEXAS A&M

OCNG 608: Physical Oceanography

Fall 2024, 2025 & 2026 (ongoing)

Graduate course, 10-25 students

This course covers a broad range of topics, including hydrostatic and geostrophic balance, Ekman transport, large scale ocean circulation, ocean stratification, mixing and surface gravity waves. Students combine mathematical and spatial problem solving with domain knowledge to further their understanding of ocean physics.

OCNG 609: Dynamical Oceanography

Spring 2022, 2024 & 2026

Graduate course, 3-7 students

This is a course about Geophysical Fluid Dynamics. In Spring 2024 I taught the course in a “flipped classroom” style, with students doing readings outside the class and problem sets during class hours with instructor guidance.

OCNG 451: Mathematical Modeling in Ocean Climate

Spring 2025

Upper level undergraduate course, approx. 45 students

This course is taken primarily by Math majors. It covers different approaches to modeling the Earth System, and includes some labs where the students run their own energy balance models. This course is usually taught by Dr Ping Chang and I taught it while he was on leave.

OCNG 310: Physical Oceanography

Spring and Fall 2023, Spring 2024

Undergraduate course, 20-45 students

This course covers a broad range of topics, including hydrostatic and geostrophic balance, Ekman transport, large scale ocean circulation, ocean stratification and surface gravity waves. As part of the course, students develop numeracy and scientific problem solving skills.

RESEARCH
MENTORING

Graduate students:

Ayantika Bhattacharjee, Texas A&M (August 2023–present)

Nicole Neumann, Texas A&M (August 2024–present)

Aoming Yu, Texas A&M (August 2024–present)

Co-advising of graduate students:

Junwei Hua, Texas A&M (June 2025–present)

Primary advisor is Dr. Jiabi Du.

External masters thesis advising:

Sanjana Satish, IISER Mohali, India (Summer 2024–Spring 2025)

Now a PhD student at University of Wisconsin, Madison.

Jones was primary advisor for Ms Satish’s Masters thesis research

Undergraduate research mentees:

Scout Jiang, Columbia University (2019–2020)

Now software developer at Amazon.

Sarah Bellefleur, Texas A&M REU (Summer 2022)

Now a data analyst.

Logan Knudsen, Texas A&M (2022–2023)

Now PhD student at the University of Maryland.

Isaiah Sutberry, Texas A&M (Spring 2022–Spring 2024)

Julianne Huesby, Texas A&M REU (Summer 2024)

William Gross, Texas A&M and University of South Carolina (Summer 2025)

Anya Moysey, Texas A&M and REU (Summer 2026)

Post-undergraduate research mentees:

Nicole Neumann, Texas A&M/Columbia University (2022–2024)

Now a PhD student at Texas A&M.

Kaila Uyeda, Texas A&M (2023–2024)

Now PhD student at UC Irvine.

Aayush Upadhyay, Texas A&M (2024–2026; Department of Computer Science)

Now CEO of a tech startup

TECHNICAL SKILLS

Scientific computer programming in MATLAB, python and FORTRAN. Experienced with git, github, cloud computing, big-data workflows and with python packages for handling large geoscience datasets, including xarray, xgcm, xmitgcm, zarr and kerchunk. Familiar with the MIT general circulation model (including the adjoint model and state estimation capabilities) and with MOM6. Recently wrote code to advect particles online in MOM6. Some experience with CESM. Proficient with LaTeX, UNIX and Microsoft Office.

INVITED TALKS

Pathways to Understanding: Lagrangian Advection in MOM6 & Tropical Atlantic Heat Transport. Los Alamos National Laboratory, May 2025.

Adventures in Atlantic Ocean Heat Transport: AMOC, gyres, and tropical upwelling. University of Texas Institute for Geophysics, January 2025.

Disentangling ocean dynamics at multiple scales: fronts vs internal waves & gyres vs. overturning. Sean Haney Memorial Symposium at Scripps Institution of Oceanography, February 2024.

Heat Transport by Overturning and Gyres: Applying New Diagnostics for Heat Transport to the CESM Large Ensemble. University of Wisconsin, Madison, December 2023.

Assessing the large parameter space of possible LGM ocean circulations. Paleoclimate Advances Webinar Series, November 2023.

Finding the transport-relevant surface velocity field using Lagrangian filtering in LLC4320. NASA GMAO, March 2022.

Separating balanced and unbalanced flow at the ocean surface using Lagrangian filtering in LLC4320. Woods Hole Oceanographic Institution, November 2021.

Understanding Subpolar North Atlantic Heat Content Variability in the CESM Large Ensemble. NASA GISS, January 2021.

Understanding ocean heat and tracer transport, both today and at the Last Glacial Maximum. University of Washington, November 2020.

The global Meridional Overturning Circulation in an idealized two-basin model. Stony Brook University, January 2019.

The global Meridional Overturning Circulation in an idealized two-basin model. Yale University, September 2018.

SERVICE &
OUTREACH

Co-Organizer, *Collapse of the Atlantic Ocean circulation: Can it? Has it? Will it? Workshop*, Slough, UK, 2026

Reviewer, *First-Time Presenter Feedback Program*, AGU Fall Meeting 2025

Dissertation committee member, *4 Masters students and 2 PhD students*, Texas A&M (Spring 2022-present)

Mentor, *Academy of Future Faculty*, one student, Texas A&M (September 2025-present)

Member, *Qualifying Examination Committee*, Department of Oceanography, Texas A&M (September 2024–present)

Member, *Recruitment and Admissions Committee*, Department of Oceanography, Texas A&M (September 2023–August 2025)

Mentor and Research Advisor, *Research Experience for Undergraduates (REU) Program*, Department of Oceanography, Texas A&M (Summer 2022 and Summer 2024)

Mentor, *Promoting Geosciences Research, Education and Success (PROGRESS) program* (Fall 2022)

Co-convenor, *PL01: Multi-scale transport of oceanographic tracers: mean flow, stirring, and mixing*, a session at the Ocean Sciences Meeting (February 2022)

Co-creator, *Pyclub: Dive into python with oceanographers*, a short course for high-schoolers (piloted Spring 2021)

Student Member, *Atmospheric and Oceanic Fluid Dynamics Committee*. American Meteorological Society (February 2014 - February 2018)

REVIEWER	Nature Nature Communications Journal of Physical Oceanography Journal of Climate Geophysical Research Letters Journal of Advances in Modeling Earth Systems Journal of Geophysical Research Geoscientific Model Development Dialogues on Climate Change
WORKING GROUPS	Member, <i>NASA Ocean AI working group</i> (November 2024–present) Member, <i>NASA SWOT velocities working group</i> (September 2025–present)